



CERTIFICATE

OF PARTICIPATION

This is to certify that **Rupak Neupane** attended the 10-day Fifth Annual Nepal AI School (ANAIIS) 2024 organized by Nepal Applied Mathematics and Informatics Institute for Research (NAAMII) at Premier International IB Continuum School, Lalitpur, Nepal from 27 December 2024 to 6 January 2025.

The 10-day (35 hrs. lectures + 12 hrs. lab sessions + 12 hrs. Professional development sessions + 10 hrs. Hackathon Project) AI School covered the following topics:

- **Day 1** : Introductory Session, Deep Learning Foundations from Geometric Viewpoint, Lab Session on Introduction to Geometric Deep Learning (GDL) and Graph Neural Networks (GNNs), Hackathon Introduction Session.
- **Day 2** : GDL: Preliminaries, Graph Convolutional Network (GCN), Graph Attention Network (GAT) and Message Passing Neural Network (MPNN), Lab Session on GCN, GAT and MPNN, Session on AI Application Design.
- **Day 3** : GDL: Advanced Topics, Introduction to Graph Transformers, Lab session on Invariant & Equivariant in MPNN, Data Engineering Mentor Session.
- **Day 4** : Self-supervised Learning and Active Learning, Foundational Models, Lab session on Self-supervised Contrastive Learning, Entrepreneurial and Team Building Session.
- **Day 5** : AI in Business and Human-AI collaboration, Vision Language Models, Lab session on Mixture of Experts (MoE) Transformer, Entrepreneurial Development and Startup Building Session.
- **Day 6** : 3D Reconstruction of Rigid and Deformable Objects, Deep Learning-Based 3D Mapping, Lab session on Contrastive Language-Image Pre-training (CLIP), Strategic insight session for Hackathon.
- **Day 7** : Gradient Descent Optimization for Deep Learning, Large Language Models (LLMs), Lab session on Fine-Tuning LLMs, UI/UX Mentor Session.
- **Day 8** : Reinforcement Learning (RL) & RL from Human Feedback, AI Risks & Safety, Lab session on Machine Learning for Agricultural Modelling, Session for Hackathon demo Preparation.
- **Day 9** : AI in Agriculture, Low Resource and Trustworthy AI in Healthcare, AI for Energy Informatics, Hackathon Presentation.
- **Day 10** : Application of AI in Healthcare, Networking and Closing Session.

OUR SPEAKERS

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| • Prof. Chetan Arora (IIT Delhi, India) | • Dr. Kamil Adamczewski (ETH Zürich, Switzerland) | • Mr. Ahmed Elhag (University Oxford, UK) |
| • Prof. Michael Bronstein (University of Oxford, UK) | • Dr. Kilian Koepsell (Caption Health/ GE Healthcare, USA) | • Mr. Federico Monti (Università della Svizzera Italiana, Switzerland) |
| • Prof. Norliza Mohd Noor (Universiti Teknologi Malaysia, Malaysia) | • Dr. Manoj Shakya (KU, Nepal) | • Mr. Haitz Sáez de Ocáriz Borde (University of Oxford, UK) |
| • Prof. Sabita Maharjan (University of Oslo, Norway/NAAMII) | • Dr. Shaifali Parashar (Liris-CNRS, France) | |
| • Dr. Ashutosh Modi (IIT Kanpur, India) | • Dr. Shiva Ram Dubey (IIT Allahabad, India) | PROFESSIONAL DEVELOPMENT SESSIONS |
| • Dr. Binod Bhattarai (University of Aberdeen, UK/NAAMII) | • Dr. Vijay Prakash Dwivedi (Stanford University, USA) | • Mr. Alok Khatri (NAAMII) |
| • Dr. Bishesh Khanal (NAAMII, Nepal) | • Dr. Vivek Choudhary (NTU, Singapore) | • Mr. Bijayan Bhattarai (Diyo.AI) |
| • Dr. Dilli Raj Paudel (Wageningen University & Research, Netherlands) | • Dr. Yash Raj Shrestha (University of Lausanne, Switzerland) | • Ms. Ishu Dharel (Woof Web Studio) |
| • Dr. François Rameau (SUNY, South Korea/ NAAMII) | | • Mr. Niradesh Dwa (Veda App) |
| | | • Mr. Manik Madhikerm (FI Soft) |
| | | • Mr. Lalit BC (Plantsat) |

Yours Sincerely,




Mr. Alok Khatri
General Co-Chair


Dr. Danda Pani Paudel
General Co-Chair


Dr. Suman Raj Bista
Program Co-Chair